Response to Reviewers

Paper No. S-28-4-002

Original title: Camerala: A Privacy-Preserving Browser-Native Framework for In-the-Wild Physiological Sensing and Task Measurement

We thank the handling editor and the reviewers for their careful reading and constructive comments. The manuscript was revised according to Reviewer 1's required Conditions 1–13. The revision intentionally narrows the claims: Camerala is now positioned as a browser-native, local-first implementation example for task-aligned physiological feature logging under limited, task-constrained home/laboratory conditions. Reviewer 2 recommended acceptance without revision.

Condition 1: The title of the manuscript should be reconsidered

Response. Thank you for this comment. The manuscript was revised to avoid implying that Camerala proposes a dedicated privacy-preserving algorithm. Following the overall revision, the title was changed to a more conservative system-integration title.

Revision. The title no longer uses "Privacy-Preserving" or an unqualified "In-the-Wild" expression. Privacy is now discussed as an architectural privacy advantage of avoiding raw-video transmission, not as a dedicated privacy-preserving method.

Key revised text. "Camerala: A Browser-Native Framework for Task-Aligned Physiological Feature Logging in Home and Laboratory Settings"

Location. Title page; Introduction; Contributions.

Condition 2: The necessity of real-time processing is not sufficiently explained

Response. Thank you for pointing this out. The manuscript now clarifies that real-time processing is treated as a design choice for local-first feature logging, rather than the only technically possible architecture.

Revision. A new paragraph was added in the Introduction explaining that local offline processing of recorded video would be possible, but would require storing privacy-sensitive raw video. The revised text explains that Camerala extracts features during task execution to avoid raw-video persistence, synchronize task events and frame features in the same browser runtime, and support online quality checks.

Key revised text. "Real-time processing is treated here as a design choice for local-first feature logging rather than as the only technically possible architecture. Recording video locally and processing it offline could preserve timestamps, but it would require storing privacy-sensitive raw video."

Location. Introduction; System Architecture; Data Management and Local-First Persistence.

Condition 3: The Research Questions are not defined clearly enough

Response. Thank you for this comment. The Research Questions were rewritten to define the tested configuration and the observation criteria more concretely.

Revision. RQ1 now asks whether, in the tested setup, Camerala can complete the deployment while locally logging synchronized task events, sensor-derived features, and quality indicators. The criteria now include planned session/trial completion, local persistence, task-frame synchronization, post-exclusion retention, and availability of the listed quality indicators.

Key revised text. “These criteria are intended only to describe operational completion under the tested configuration; they do not demonstrate general deployability across users, devices, or environments.”

Location. Introduction, Research Questions.

Condition 4: It is unclear what the “between-environment differences” in RQ2 refer to

Response. Thank you for this important clarification. RQ2 was redefined to concern descriptive differences in measurement-quality indicators and sensor-derived estimates, not psychological, cognitive, motivational, or physiological effects.

Revision. The Abstract, RQ2, Results, Discussion, and Conclusion were revised to state that the observed laboratory/home differences are deployment-specific sensor-derived observations. The manuscript now explicitly states that the study did not include a reference physiological sensor, multiple participants, or controlled separation of environmental and task-related factors.

Key revised text. “What descriptive differences in measurement-quality indicators and sensor-derived estimates are observed between the laboratory and home deployment contexts?”

Location. Abstract; Introduction, Research Questions; Results; Discussion; Conclusion.

Condition 5: The novelty and positioning of the study are insufficiently justified

Response. Thank you for identifying the relevant prior work. The Related Work section was expanded, and the novelty claim was repositioned.

Revision. The Related Work section now discusses relevant web, edge, and privacy-oriented rPPG systems and public tools, including Gupta and Etemad, Ayesha et al., RhythmEdge, Labvanced rPPG, FacePhys-Demo, and Di Lernia et al. The manuscript explicitly states that browser-based rPPG functionality, local rPPG processing alone, edge-based rPPG, and privacy preservation as an algorithm are not the novelty of Camerala. Camerala’s contribution is now framed as the integration of psychophysical task execution, on-device rPPG/FaceMesh-derived feature extraction, measurement-quality indicators, and timestamped local persistence within one browser-native runtime.

Key revised text. “The contribution of Camerala is instead the system-level integration of psychophysical task execution, on-device rPPG/FaceMesh-derived feature extraction, frame-level measurement-quality indicators, and timestamped local persistence within a unified browser-native runtime.”

Location. Related Work; Contributions; System Architecture; References.

Condition 6: Unclear claim of robustness against rigid head motion

Response. Thank you for this comment. Unsupported robustness claims were removed.

Revision. The revised manuscript states that Camerala does not attempt to correct rPPG degradation caused by motion or illumination changes. Head motion and luminance variation are now presented as logged quality indicators for inspection and filtering.

Key revised text. "Camerala does not attempt to correct rPPG degradation caused by motion or illumination changes. Instead, it records motion-related and luminance-related indicators so that data quality can be inspected and, if necessary, filtered during analysis."

Location. System Architecture, Feature Extraction Logic; Discussion.

Condition 7: Insufficient description of the concrete usage scenario

Response. Thank you for the comment. The intended usage scenario is now stated at the beginning of the paper and again in the Methodology.

Revision. The manuscript now defines the target use case as a remote or home-based psychophysical task performed while seated in front of a laptop, with synchronized browser-based logging of behavioral events and webcam-derived features.

Key revised text. "The target usage scenario is a remote or home-based psychophysical task performed while seated in front of a laptop, with browser-based logging of behavioral events and webcam-derived features."

Location. Introduction; Methodology, Target Usage Scenario and Exploratory Deployment.

Condition 8: Definition of "in-the-wild" and "naturalistic physiological sensing" are not clear

Response. Thank you for this point. The manuscript was revised to avoid giving the impression that Camerala was evaluated for unconstrained daily-life physiological sensing.

Revision. Unqualified "in-the-wild" and "naturalistic" language was removed or replaced with "task-constrained home/laboratory deployment." The Methodology now explicitly states that the deployment does not represent continuous sensing during daily activities such as walking, cooking, conversation, or night-time use.

Key revised text. "This is a task-constrained home/laboratory deployment, not continuous physiological sensing during ordinary daily activities such as walking, cooking, conversation, or night-time use."

Location. Abstract; Introduction; Methodology; Discussion; Conclusion.

Condition 9: The description of the system hardware is insufficient for reproducibility

Response. Thank you for this comment. A hardware and software configuration table was added.

Revision. The hardware/software table was revised using author-confirmed information: PC model, CPU, GPU, RAM, storage, OS, browsers, built-in webcam, requested camera constraints, browser-negotiated capture behavior, display resolution/refresh rate, fixed screen brightness, viewing distance, and default camera-control behavior.

Key revised text. "Camera capture behavior: Browser-negotiated capture under the requested constraints; no manual exposure or white-balance constraints were applied."

Location. Methodology, Hardware and Software Configuration; Table 2.

Condition 10: The description of the experimental environments is insufficient

Response. Thank you for the detailed request. An environment and lighting conditions table was added.

Revision. The previous lux-based description was removed because lighting is reported as camera-derived brightness rather than external lux. The revised section reports laboratory/home room type, daytime sessions, windows, closed curtains, general artificial indoor lighting, fixed screen brightness, and camera-derived brightness ranges.

Key revised text. "Because the deployment used camera-derived brightness values rather than external light-meter measurements, this paper reports brightness as a camera-derived measurement-quality indicator."

Location. Methodology, Deployment Environment and Lighting Conditions; Table 3.

Condition 11: The validity of the cognitive framing manipulation is questionable

Response. Thank you for this important point. The manuscript no longer interprets the instruction blocks as inducing evaluation pressure, reward motivation, cognitive effects, or motivational effects.

Revision. The terminology was changed from cognitive framing manipulation to instructional task contexts. The participant/autoethnography section now states that the author-participant knew the instructions were experimental.

Key revised text. "They are used only as different instructional task contexts for demonstrating synchronized logging across task conditions."

Location. Participant section; Experimental Procedure and Instructional Task Contexts; Results.

Condition 12: The central claim in the Discussion should be reconsidered

Response. Thank you for this comment. The Discussion was rewritten around design requirements and safeguards for future Camerala deployments.

Revision. The revised Discussion focuses on browser-native local-first logging, quality indicators for task-constrained home/laboratory deployment, interpretation under environmental heterogeneity, and design safeguards. The previous central claim about laboratory-only validation overestimating home-use signal stability was removed as the organizing claim.

Key revised text. "The results do not establish general deployability, robustness, or physiological/cognitive effects. Instead, they motivate design requirements and safeguards for future local-first deployments."

Location. Discussion.

Condition 13: Several claims are not sufficiently supported and should be weakened

Response. Thank you for this guidance. The Abstract, Results, Discussion, Limitations, and Conclusion were revised to substantially weaken unsupported claims.

Revision. The manuscript no longer states that the study "establishes" a design pattern. The claims about feasibility, naturalistic deployment, tracking confidence, and psychological and physiological interpretation were narrowed. The face-tracking threshold ($Conf < 0.8$) is explained as a conservative operational FaceMesh threshold, not an rPPG quality threshold. Face-tracking availability, including the 96.5% home value after filtering, is no longer used as strong evidence of rPPG quality or general deployment feasibility.

Key revised text. "Therefore, face-tracking availability after filtering should be interpreted only as a FaceMesh/logging quality indicator, not as direct evidence of rPPG signal quality, landmark accuracy, or deployment feasibility."

Location. Abstract; Results; Discussion; Limitations; Conclusion.

Reviewer 1 Other Comments

Response. Thank you for the additional comments on figure and table readability.

Revision. Table 1 was reformatted with clearer column widths. The previous longitudinal and task-result figures contained overstrong embedded labels, so they were removed from the revised manuscript rather than retained. The remaining results presentation relies on conservative text and tables without p-values, significance marks, or causal wording.

Location. Table 1; Results; removed previous Figure 2, Figure 3, and Figure 4 from the revised manuscript.